Parametric Equations

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.03

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Eliminating the parameter**

Consider the curve $x = 2t + 1$, $y = t^2$.

a Eliminate the parameter to write y as a function of x. $t = \frac{x-1}{2}$, so $y = \frac{(x-1)^2}{4}$.

b Identify the curve. A parabola with vertex $(1, 0)$ opening upward.

(a) $t = \frac{x-1}{2}$, so $y = \frac{(x-1)^2}{4}$.; (b) A parabola with vertex $(1, 0)$ opening upward.

2 **[Foundational]** • **Target: Eliminating the parameter**

Eliminate the parameter for $x = 3\cos t$, $y = 3\sin t$, $0 \le t < 2\pi$, and describe the curve.

$x^2 + y^2 = 9\cos^2 t + 9\sin^2 t = 9$: a circle of radius 3 centered at the origin, traced counterclockwise.

3 **[Foundational]** • **Target: Eliminating the parameter**

For the line $x = t - 2$, $y = 2t + 3$:

a Find the points corresponding to $t = 0$ and $t = 2$. $t = 0$: $(-2, 3)$; $t = 2$: $(0, 7)$.

b Eliminate the parameter to write the Cartesian equation. $t = x + 2$, so $y = 2x + 7$.

(a) $t = 0$: $(-2, 3)$; $t = 2$: $(0, 7)$.; (b) $t = x + 2$, so $y = 2x + 7$.

4 **[Foundational]** • **Target: Projectile motion**

A projectile follows $x = 40t$, $y = -5t^2 + 30t$ (metres, seconds).

a When does the projectile land ($y = 0$, $t > 0$)? $-5t(t - 6) = 0$, so $t = 6$ s.

b Find the horizontal range. $x(6) = 240$ m.

c Find the maximum height. Peak at $t = 3$: $y(3) = -45 + 90 = 45$ m.

(a) $-5t(t - 6) = 0$, so $t = 6$ s.; (b) $x(6) = 240$ m.; (c) Peak at $t = 3$: $y(3) = -45 + 90 = 45$ m.

5 **[Foundational]** • **Target: Projectile motion**

A second projectile is launched with $x = 25t$, $y = -5t^2 + 40t$. Find its time of flight and maximum height.

Land: $-5t(t - 8) = 0$, so $t = 8$ s. Peak at $t = 4$: $y(4) = -80 + 160 = 80$ m.

© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: Ellipses and other parametric curves

Eliminate the parameter for $x = 4\cos t$, $y = 2\sin t$, and identify the resulting curve.

$\frac{x}{4} = \cos t$, $\frac{y}{2} = \sin t$: $\frac{x^2}{16} + \frac{y^2}{4} = 1$ — an ellipse with semi-axes 4 and 2.

7 [Foundational] • Target: Ellipses and other parametric curves

Eliminate the parameter for $x = t^2 - 1$, $y = t^3$ by expressing t in terms of x where possible, or by finding a relation between x and y directly.

From $x = t^2 - 1$: $t^2 = x + 1$. Squaring $y = t^3$: $y^2 = t^6 = (t^2)^3 = (x + 1)^3$. Relation: $y^2 = (x + 1)^3$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Finding the parametric derivative

For the curve $x = t^2$, $y = 4t$, find $\frac{dy}{dx}$ in terms of t using $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$.

$\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 4$: $\frac{dy}{dx} = \frac{4}{2t} = \frac{2}{t}$ (for $t \neq 0$).

9 [Intermediate] • Target: Finding the parametric derivative

For the curve $x = \cos t$, $y = \sin t$, find the slope of the tangent line at $t = \frac{\pi}{4}$.

$\frac{dx}{dt} = -\sin t$, $\frac{dy}{dt} = \cos t$: $\frac{dy}{dx} = -\cot t$. At $t = \frac{\pi}{4}$: slope $= -\cot \frac{\pi}{4} = -1$.

10 [Intermediate] • Target: More parametric practice

A Ferris wheel of radius 10 m has its center 12 m above the ground and completes one revolution every 40 seconds. Write parametric equations for the height y of a rider above the ground as a function of time t (seconds), assuming the rider starts at the bottom of the wheel.

Angular speed $\omega = \frac{2\pi}{40} = \frac{\pi}{20}$. Starting at the bottom means the vertical position begins at a minimum, matching $-\cos$: $y(t) = 12 - 10\cos\left(\frac{\pi}{20}t\right)$.

11 [Intermediate] • Target: More parametric practice

Two particles travel along paths $x_1 = t,\ y_1 = t^2$ and $x_2 = 4 - t,\ y_2 = 3t - 2$, sharing the same parameter t (time). Find the value of t at which they collide, and the point of collision.

Need $x_1 = x_2$: $t = 4 - t$, so $t = 2$. Check $y_1 = y_2$ at $t = 2$: $y_1 = 2^2 = 4$; $y_2 = 3(2) - 2 = 4$. Both match, confirming a genuine collision at $t = 2$, at the point $(2, 4)$.

© ClipSAT — clipsat.org | All Rights Reserved

12 [Intermediate] • Target: Graphing a parametric curve after elimination

For $x = t$, $y = t^2 - 2$ (with t ranging over all reals), the eliminated Cartesian form $y = x^2 - 2$ is shown. Use the graph to describe the direction the curve is traced as t increases.

As t increases from $-\infty$ to ∞, the curve is traced from the upper-left, down through the vertex $(0, -2)$, then up to the upper-right — left to right along the parabola.

13 [Intermediate] • Target: Restricting the parameter domain

The curve $x = \cos t$, $y = \sin t$ for $t \in [0, \pi]$ traces only part of the unit circle. Describe which part, and state the domain and range restrictions this implies on x and y.

For $t \in [0, \pi]$, $y = \sin t \geq 0$, so only the UPPER half of the unit circle is traced (from $(1, 0)$ counterclockwise to $(-1, 0)$). This restricts $y \geq 0$ while x still ranges over $[-1, 1]$.

14 [Intermediate] • Target: Parametrizing a line segment

Write parametric equations for the line segment from $A(2, -1)$ to $B(6, 5)$, using $t \in [0, 1]$ (so $t = 0$ gives A and $t = 1$ gives B).

$x(t) = 2 + 4t$, $y(t) = -1 + 6t$ for $t \in [0, 1]$ (using $x = x_1 + t(x_2 - x_1)$, $y = y_1 + t(y_2 - y_1)$).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Parametrizing a line segment • Cross-Lesson: 1.03 Average Rate of Change

Using the parametrization above, find the point at $t = 0.25$ (one-quarter of the way from A to B).

$x(0.25) = 2 + 1 = 3$; $y(0.25) = -1 + 1.5 = 0.5$. Point: $(3, 0.5)$.

16 [Challenging] • Target: Parametric speed and motion review • Cross-Lesson: 2.01 Right-Triangle Trigonometry

A particle's position is $x(t) = 3t$, $y(t) = 4t$ for $t \geq 0$ (measured in meters, seconds). Find the particle's speed, and confirm it matches $\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$.

$\frac{dx}{dt} = 3$, $\frac{dy}{dt} = 4$. Speed $= \sqrt{9 + 16} = 5$ m/s (constant, matching straight-line motion at constant velocity).

17 [Challenging] • Target: Parametrizing a line segment • Cross-Lesson: 2.02 Radian Measure and Arc Length

Eliminate the parameter t from the parametric curve $x(t) = 3 + 4\cos t$, $y(t) = -2 + 5\sin t$ for $0 \leq t \leq 2\pi$. Identify the conic section, its center, and its vertices.

Step 1: Isolate $\cos t$ and $\sin t$: $\cos t = \frac{x-3}{4}$, $\sin t = \frac{y+2}{5}$. Step 2: Apply the Pythagorean identity $\cos^2 t + \sin^2 t = 1$: $\mathbf{\frac{(x-3)^2}{16} + \frac{(y+2)^2}{25} = 1}$. This is a vertical ellipse centered at $\mathbf{(3, -2)}$ with major-axis vertices at $\mathbf{(3, -7)}$ and $\mathbf{(3, 3)}$.

© ClipSAT — clipsat.org | All Rights Reserved

18 [Challenging] • Target: Parametric speed and motion review • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Find the standard equation $Ax + By = C$ of the line perpendicular to $3x - 4y = 12$ that passes through the midpoint of the line segment joining $P(-2, 7)$ and $Q(6, 1)$.

(a) Midpoint M: $M = \left(\frac{-2+6}{2}, \frac{7+1}{2}\right) = \left(\frac{4}{2}, \frac{8}{2}\right) = (2, 4)$. (b) Perpendicular slope: Rewrite the given line in slope-intercept form: $4y = 3x - 12 \Rightarrow y = \frac{3}{4}x - 3$. The slope is $m = \frac{3}{4}$. The perpendicular slope is the negative reciprocal: $m_\perp = -\frac{4}{3}$. (c) Line equation: Using point-slope form with point $(2, 4)$.

19 [Challenging] • Target: Eliminating the parameter • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Find the exact area of the region enclosed between the curve $y = 4x - x^2$ and the line $y = x$.

Find points of intersection: $4x - x^2 = x \Rightarrow 3x - x^2 = 0 \Rightarrow x(3 - x) = 0$ The curves intersect at $x = 0$ and $x = 3$. Between 0 and 3, the curve $y = 4x - x^2$ lies above the line $y = x$. The enclosed area is: $A = \int_0^3 [(4x - x^2) - x]\,dx = \int_0^3 (3x - x^2)\,dx$ Integrate: $A = \left[\frac{3x^2}{2} - \frac{x^3}{3}\right]_0^3 = \left(\frac{3(9)}{2} - \frac{27}{3}\right) - 0 = \frac{27}{2} - 9 = \frac{27-18}{2} = \frac{9}{2} = 4.5$

20 [Challenging] • Target: Projectile motion • Cross-Lesson: 3.01 The Unit Circle and Exact Values

A curve has equation $y = 2x^3 - 9x^2 + 12x + 1$. (a) Find the coordinates of both stationary points. (b) Determine the nature of each stationary point using the second derivative $\frac{d^2y}{dx^2}$.

(a) Find $\frac{dy}{dx}$: $\frac{dy}{dx} = 6x^2 - 18x + 12$ Set $\frac{dy}{dx} = 0$: $6(x^2 - 3x + 2) = 0 \Rightarrow 6(x - 1)(x - 2) = 0$, giving $x = 1$ and $x = 2$. When $x = 1$: $y = 2(1) - 9(1) + 12(1) + 1 = 6$. Point is $(1, 6)$. When $x = 2$: $y = 2(8) - 9(4) + 12(2) + 1 = 16 - 36 + 24 + 1 = 5$. Point is $(2, 5)$. (b) Second derivative: $\frac{d^2y}{dx^2} = 12x - 18$ At $x = 1$: $\frac{d^2y}{dx^2} = 12(1) - 18 = -60$, so $(2, 5)$ is a local minimum.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Eliminating the parameter

A particle of mass 6 kg moving at 6 m/s collides and sticks to a particle of mass 6 kg moving at 0 m/s (same line of motion). Find the common velocity after collision.

a. 2

b. 4

c. 6

d. **3** ✓

By conservation of momentum: $m_1u_1 + m_2u_2 = (m_1 + m_2)v \Rightarrow (6)(6) + (6)(0) = (12)v \Rightarrow v = 3$ m/s

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • Target: Projectile motion

A particle moves in a straight line such that its displacement s metres from a fixed origin after t seconds is given by the equation $s = t^3 - 6t^2 + 9t + 2$. The velocity of the particle is given by $v = \frac{ds}{dt}$. Find the value of t for which the particle is instantaneously at rest and then determine the displacement of the particle at that instant.

a. t=1 s, s=6 m

b. t=2 s, s=4 m

c. **t=3 s, s=2 m ✓**

d. t=1 s, s=2 m

The displacement is given by $s = t^3 - 6t^2 + 9t + 2$. To find the velocity, we differentiate the displacement with respect to time: $v = \frac{ds}{dt} = \frac{d}{dt}(t^3 - 6t^2 + 9t + 2) = 3t^2 - 12t + 9$. The particle is instantaneously at rest when its velocity is zero, i.e., $v = 0$. So, we set the velocity equation to zero: $3t^2 - 12t + 9 = 0$. Divide by 3: $t^2 - 4t + 3 = 0$. Factorize the quadratic equation: $(t - 1)(t - 3) = 0$. This gives two possible values for t: $t = 1$ or $t = 3$. The question asks for 'the.

23 [Foundational] • Target: Ellipses and other parametric curves

The region R in the first quadrant is bounded by the curve $y = \frac{6}{\sqrt{3x+4}}$, the coordinate axes, and the vertical line $x = k$, where $k > 0$. Given that the total area of R is 8, and the vertical line $x = c$ (where $0 < c < k$) divides R into two regions of equal area, find the exact value of c.

a. $\frac{5}{3}$ ✓

b. 2

c. $\frac{7}{3}$

d. $\frac{13}{27}$

Step 1: Integrate the function using the reverse chain rule. $\int 6(3x+4)^{-\frac{1}{2}}\,dx = \frac{6}{\frac{1}{2}\cdot 3}(3x+4)^{\frac{1}{2}} = 4\sqrt{3x+4}$

Step 2: Find the parameter k using the given total area of 8.

$\text{Area} = \int_0^k \frac{6}{\sqrt{3x+4}}\,dx = \left[4\sqrt{3x+4}\right]_0^k = 4\sqrt{3k+4} - 4\sqrt{4} = 4\sqrt{3k+4} - 8$ Setting this equal to 8:

$4\sqrt{3k+4} - 8 = 8 \Rightarrow 4\sqrt{3k+4} = 16 \Rightarrow \sqrt{3k+4} = 4$ Squaring both.

© ClipSAT — clipsat.org | All Rights Reserved

24 [Foundational] • Target: Finding the parametric derivative

The curve C has equation $y = \frac{ax+6}{x^2+3}$ where a is a constant. Given that C has a stationary point at $x = 1$, find the coordinates of the other stationary point of C.

a. $(-3, -1)$ ✓

b. $(-3, 1)$

c. $(3, 1)$

d. $(-3, -2)$

To find the unknown parameter a, differentiate y with respect to x using the quotient rule: $\frac{dy}{dx} = \frac{a(x^2+3)-(ax+6)(2x)}{(x^2+3)^2} = \frac{3a-12x-ax^2}{(x^2+3)^2}$ At a stationary point, $\frac{dy}{dx} = 0$, which requires the numerator to be zero: $3a - 12x - ax^2 = 0$ Given that $x = 1$ is a stationary point: $3a - 12(1) - a(1)^2 = 0 \Rightarrow 2a - 12 = 0 \Rightarrow a = 6$ Substitute $a = 6$ back into the numerator of the derivative.

25 [Foundational] • Target: More parametric practice

A storage bin contains red, blue, and green blocks. The ratio of red blocks to blue blocks is $3 : 5$. After 12 green blocks are added to the bin, with no other blocks added or removed, the total number of blocks in the bin is 60, and the probability of randomly selecting a green block from the bin is $\frac{1}{3}$. How many red blocks were in the bin originally?

a. 12

b. **15** ✓

c. 18

d. 25

1. Find the total number of green blocks after the addition: Since the new total is 60 and the probability of selecting a green block is $\frac{1}{3}$, the number of green blocks is: $60 \times \frac{1}{3} = 20$ 2. Determine the number of red and blue blocks: The remaining blocks in the bin are red and blue. Since green blocks account for 20 of the 60 blocks, the combined total of red and blue blocks is: $60 - 20 = 40$ (Alternatively, originally there were $20 - 12 = 8$ green blocks, and the original total was.

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Graphing a parametric curve after elimination

In the diagram below, line segment DE is parallel to BC. Given that AD = 6, DB = 4, and DE = 12, find the length of BC.

a. 18

b. **20** ✓

c. 30

d. 15

Since DE is parallel to BC, triangle ADE is similar to triangle ABC (AA similarity: $\angle DAE = \angle BAC$ as they are the same angle, and $\angle ADE = \angle ABC$ as corresponding angles when DE || BC and AB is a transversal). Because the triangles are similar, the ratio of corresponding sides is equal: $\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}$ We are given AD = 6 and DB = 4. Therefore, AB = AD + DB = 6 + 4 = 10. Using the ratio of similarity: $\frac{AD}{AB} = \frac{DE}{BC}$ Substitute the known values.

27 [Foundational] • Target: Restricting the parameter domain

A particle moves along a straight line. Its velocity, v m/s, at time t seconds is given by $v = 3t - 2$. Find the displacement of the particle during the first 4 seconds of its motion.

a. 16 m

b. **20 m** ✓

c. 24 m

d. 28 m

The displacement of the particle can be found by integrating the velocity function with respect to time. The displacement s is given by: $s = \int v\,dt$ In this case, $v = 3t - 2$. $s = \int (3t - 2)\,dt$ Integrating, we get: $s = \frac{3}{2}t^2 - 2t + C$ To find the displacement during the first 4 seconds, we evaluate the definite integral from $t = 0$ to $t = 4$. We can assume $C = 0$ for calculating the change in displacement from $t = 0$. $s(4) - s(0) = \left[\frac{3}{2}t^2 - 2t\right]_0^4$.

28 [Foundational] • Target: Parametrizing a line segment

A particle has initial velocity $u = 3$ m/s and constant acceleration $a = 3$ m/s^2. Find its speed after travelling a displacement $s = 11$ m.

a. $\sqrt{79}$

b. $\sqrt{71}$

c. 75

d. $\sqrt{75}$ ✓

$v^2 = u^2 + 2as = (3)^2 + 2(3)(11) = 75 \Rightarrow v = \sqrt{75}$ m/s

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

29 [Intermediate] • Target: Parametric speed and motion review

Consider the function $f(x) = x^3 - 6x^2 + 9x + 1$. A line tangent to the graph of $f(x)$ at $x = 1$ intersects the graph at another point P. Determine the x-coordinate of point P.

a. **4** ✓

b. 5

c. 3

d. 2

First, find the derivative of $f(x)$ to determine the slope of the tangent line: $f'(x) = 3x^2 - 12x + 9$. The slope of the tangent line at $x = 1$ is $f'(1) = 3(1)^2 - 12(1) + 9 = 3 - 12 + 9 = 0$. The y-coordinate of the point of tangency is $f(1) = (1)^3 - 6(1)^2 + 9(1) + 1 = 1 - 6 + 9 + 1 = 5$. The equation of the tangent line is $y - y_1 = m(x - x_1)$, which becomes $y - 5 = 0(x - 1)$, so $y = 5$. To find the intersection points, set $f(x)$ equal to the tangent line's equation: $x^3 - 6x^2 + 9x + 1 = 5$. Rearra.

30 [Intermediate] • Target: Eliminating the parameter

The diagram shows the graphs of two linear equations. The point of intersection of the two lines represents the solution to the system of simultaneous equations. Find the coordinates of the point of intersection.

a. (1, 2)

b. **(2, 1)** ✓

c. (3, 0)

d. (0, 3)

The point of intersection of two lines is the solution to the system of simultaneous equations that represent these lines. By visually inspecting the graph, we can identify the coordinates where the two lines cross. The first line appears to pass through the points (0, 3) and (3, 0). The second line appears to pass through the points (0, 1) and (2, 3). Let's find the equations of the lines. For the first line: Gradient $m_1 = \frac{0-3}{3-0} = \frac{-3}{3} = -1$. Using the point-slope form with (3, 0).

31 [Intermediate] • Target: Projectile motion

The closed interval $[12, 48]$ is shown on the number line below. How many integers x in this interval satisfy the linear congruence $2x \equiv 1 (\text{mod } 5)$?

a. 6

b. 7

c. **8** ✓

d. 9

Step 1: Solve the linear congruence $2x \equiv 1 (\text{mod } 5)$. Since $\gcd(2, 5) = 1$, the modular inverse of 2 modulo 5 exists. We observe that $2 \times 3 = 6 \equiv 1 (\text{mod } 5)$, so the inverse is 3. Multiplying both sides by 3: $x \equiv 3 (\text{mod } 5)$ Thus, the solutions are integers of the form $x = 5k + 3$ for some integer k. Step 2: Restrict to the closed interval $[12, 48]$: $12 \leq 5k + 3 \leq 48$ Subtract 3 from each part: $9 \leq 5k \leq 45$ Divide by 5.

© ClipSAT — clipsat.org | All Rights Reserved

32 [Intermediate] • Target: Ellipses and other parametric curves

The graph of the exponential function $f(x) = a \cdot \log_3(x - h) + k$ is shown below. The function has a vertical asymptote at $x = 3$ and passes through the points $(4, 5)$ and $(12, 7)$. Determine the value of the parameter a.

a. **1 ✓**

b. 2

c. 1/2

d. 4

The general form of the logarithmic function is $f(x) = a \cdot \log_b(x - h) + k$. The vertical asymptote occurs at $x = h$, so we are given that $h = 3$. Thus, the function can be written as $f(x) = a \cdot \log_3(x - 3) + k$. We are given two points on the graph: $(4, 5)$ and $(12, 7)$. Substituting these points into the function, we get a system of two equations: 1) $5 = a \cdot \log_3(4 - 3) + k$ $5 = a \cdot \log_3(1) + k$ Since $\log_3(1) = 0$, this simplifies to $5 = k$. 2) $7 = a \cdot \log_3(12 - 3) + k$.

33 [Intermediate] • Target: Finding the parametric derivative

A curve is given by the equation $y = x^3 - 3x^2 + 2x$. Find the gradient of the curve at $x = 2$.

a. 1

b. **-2 ✓**

c. 2

d. 0

To find the gradient of the curve, we need to find the first derivative of the equation with respect to x. The derivative of $y = x^3 - 3x^2 + 2x$ is $\frac{dy}{dx} = 3x^2 - 6x + 2$. To find the gradient at $x = 2$, substitute $x = 2$ into the derivative: $\frac{dy}{dx}|_{x=2} = 3(2)^2 - 6(2) + 2 = 3(4) - 12 + 2 = 12 - 12 + 2 = 2$.

34 [Intermediate] • Target: More parametric practice

Find the equation of the line through $(0, -7)$ and $(-4, -5)$, in the form $y = mx + c$.

a. y=-1/2x+7

b. **y=-1/2x-7 ✓**

c. y=-1/2x-6

d. y=1/2x-7

Using $y - y_1 = m(x - x_1)$: $y + 7 = -1/2(x - 0)$, which rearranges to $y = -1/2x - 7$.

© ClipSAT — clipsat.org | All Rights Reserved

35 [Intermediate] • Target: Graphing a parametric curve after elimination

Find the slope of the line through $(-4, -2)$ and $(-7, -4)$.

a. **2/3** ✓

b. -4/3

c. 8/3

d. 13/6

$m = \frac{-4+2}{-7+4} = 2/3$.

36 [Intermediate] • Target: Restricting the parameter domain

A particle of mass 2 kg is acted upon by a constant force $\mathbf{F}$. At time $t = 0$, the particle is at the origin $(0, 0)$ and has velocity $\mathbf{v} = (3\mathbf{i} + 4\mathbf{j})$ m/s. At time $t = 2$ s, the particle's velocity is $\mathbf{v} = (7\mathbf{i} + 12\mathbf{j})$ m/s. Determine the magnitude of the constant force $\mathbf{F}$ acting on the particle.

a. $\frac{5\sqrt{17}}{2}$ N

b. $5\sqrt{17}$ N ✓

c. $\frac{5\sqrt{2}}{2}$ N

d. $5\sqrt{2}$ N

We are given the mass of the particle $m = 2$ kg. The initial velocity at $t = 0$ is $\mathbf{v}_0 = (2\mathbf{i} + 8\mathbf{j})$ m/s. The final velocity at $t = 2$ s is $\mathbf{v}_f = (7\mathbf{i} + 28\mathbf{j})$ m/s. The change in velocity is $\Delta\mathbf{v} = \mathbf{v}_f - \mathbf{v}_0$.

$\Delta\mathbf{v} = (7\mathbf{i} + 28\mathbf{j}) - (2\mathbf{i} + 8\mathbf{j}) = (7-2)\mathbf{i} + (28-8)\mathbf{j} = 5\mathbf{i} + 20\mathbf{j}$ m/s. The time interval i.

37 [Intermediate] • Target: Parametrizing a line segment

In the standard (x, y) coordinate plane, the line $y = mx$ (where $m > 0$) is tangent to the circle $(x - 10)^2 + y^2 = 36$. What is the value of m?

a. $\frac{3}{5}$

b. $\frac{3}{4}$ ✓

c. $\frac{4}{5}$

d. $\frac{4}{3}$

We are given the circle $(x - 10)^2 + y^2 = 36$, which has its center at $C(10, 0)$ and radius $r = \sqrt{36} = 6$. **Method 1: Geometric Approach (Right Triangle)** Let $O(0, 0)$ be the origin and let T be the point of tangency in the first quadrant. A radius drawn to the point of tangency is perpendicular to the tangent line, so ΔOTC is a right triangle with right angle at T. - Hypotenuse: $OC = 10$ - Leg (radius): $CT = 6$ By the Pythagorean theorem.

© ClipSAT — clipsat.org | All Rights Reserved

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Parametric speed and motion review • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

A factory produces light bulbs. The probability that a randomly selected light bulb is defective is p. The number of defective light bulbs in a random sample of n bulbs follows a binomial distribution $X \sim B(n, p)$. It is known that the mean number of defective bulbs in a sample of $n = 25$ is 2. \\ (i) Calculate the value of p. \\ (ii) For a sample of $n = 50$ bulbs, calculate the probability of finding exactly k defective bulbs, where k is the smallest integer such that $P(X \le k) > 0.9$. \\ (iii) If a sample of $n = 50$ bulbs is taken, find the smallest value of n' such that the probability of observing at least one defective bulb is greater than 0.999 .

a. 0.04

b. **0.08** ✓

c. 0.12

d. 0.16

Part (i): For a binomial distribution $X \sim B(n, p)$, the mean is given by $E(X) = np$. \\ We are given $n = 25$ and $E(X) = 2$. \\ So, $25p = 2$, which gives $p = \frac{2}{25} = 0.08$. \\ Part (ii): Now, $n = 50$ and $p = 0.08$. The number of defective bulbs follows $X \sim B(50, 0.08)$. \\ We need to find the smallest integer k such that $P(X \le k) > 0.9$. \\ We can use a binomial cumulative distribution table or a calculator. \\ Let's calculate a few values: $P(X \le 3) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$ Using.

39 [Challenging] • Target: Eliminating the parameter • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Consider the rational function $f(x) = \frac{x^2 - 2x + c}{x^2 + 2x + c}$ where c is a real constant. If the domain of f is the set of all real numbers and the maximum value of $f(x)$ is 4, what is the value of c?

a. $\frac{25}{9}$ ✓

b. $\frac{9}{4}$

c. 3

d. $\frac{16}{5}$

We are given the rational function $f(x) = \frac{x^2 - 2x + c}{x^2 + 2x + c}$. **Step 1: Domain constraint** For the domain of f to be all real numbers, the denominator $x^2 + 2x + c$ must have no real roots. Therefore, its discriminant must be strictly negative: $\Delta_{\text{denom}} = 2^2 - 4(1)(c) = 4 - 4c1$ **Step 2: Range and maximum value via discriminant method** Let $y = f(x)$. Because $x^2 + 2x + c > 0$ for all $x \in \mathbb{R}$, we can multiply through by the denominator.

© ClipSAT — clipsat.org | All Rights Reserved

40 [Challenging] • Target: Projectile motion • Cross-Lesson: 3.01 The Unit Circle and Exact Values

In the standard (x, y) coordinate plane, line L is perpendicular to the line $3x + 4y = 15$ and passes through the point $(3, 1)$. Line L intersects the line $2x + y = 17$ at the point (p, q). What is the value of $p + q$?

a. 5

b. 6

c. 8

d. **11** ✓

Step 1: Find the slope of the given line $3x + 4y = 15$. Rewriting in slope-intercept form gives: $4y = -3x + 15 \Rightarrow y = -\frac{3}{4}x + \frac{15}{4}$ Thus, the slope of this line is $m_1 = -\frac{3}{4}$. Step 2: Determine the slope of line L. Since line L is perpendicular to the given line, its slope m_L is the negative reciprocal of m_1: $m_L = -\frac{1}{-\frac{3}{4}} = \frac{4}{3}$ Step 3: Write the equation of line L. Line L passes through $(3, 1)$ with slope $\frac{4}{3}$.

41 [Challenging] • Target: Ellipses and other parametric curves • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

A curve has equation $y = x^2e^{3x}$. Find the gradient of the curve at the point where $x = 1$.

a. $5e^3$ ✓

b. $6e^3$

c. $3e^3$

d. $2e^3$

To find the gradient of the curve, differentiate $y = x^2e^{3x}$ with respect to x using the product rule: $\frac{d}{dx}[uv] = u\frac{dv}{dx} + v\frac{du}{dx}$ Let $u = x^2$ and $v = e^{3x}$. Then $\frac{du}{dx} = 2x$ and, by the chain rule, $\frac{dv}{dx} = 3e^{3x}$. Applying the product rule: $\frac{dy}{dx} = x^2(3e^{3x}) + 2x(e^{3x}) = (3x^2 + 2x)e^{3x}$ Substitute $x = 1$.

42 [Challenging] • Target: Finding the parametric derivative • Cross-Lesson: 3.03 Sum and Difference Formulas

A particle of mass 8 kg is moving under the action of a net force $\mathbf{F}$ N. The acceleration of the particle, $\mathbf{a}$ m/s^2, is given by the vector $\mathbf{a} = (2t - 4; t^2 - 9)$ for $t \geq 0$, where t is the time in seconds. The particle is initially at rest at the origin.

a. 8 N

b. **16 N** ✓

c. 24 N

d. 32 N

We are given the acceleration vector $\mathbf{a} = (2t - 4; t^2 - 9)$ and the mass of the particle $m = 8$ kg. By Newton's Second Law, the net force $\mathbf{F} = m\mathbf{a}$. Therefore, $\mathbf{F} = 8(2t - 4; t^2 - 9) = (16t - 32; 8t^2 - 72)$. The question asks for the magnitude of the net force at $t = 2$ seconds. Substituting $t = 2$ into the force vector.

© ClipSAT — clipsat.org | All Rights Reserved

43 [Challenging] • Target: More parametric practice • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

The random variable X has a Poisson distribution with parameter λ, where $\lambda > 0$. Given that $P(X = 3) = P(X = 4)$, find the value of λ.

a. **4** ✓

b. 3

c. 12

d. $\frac{4}{3}$

For a random variable $X \sim Po(\lambda)$, the probability mass function is given by: $P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}, \quad x = 0, 1, 2, \ldots$ Given $P(X = 3) = P(X = 4)$, we substitute $x = 3$ and $x = 4$: $\frac{e^{-\lambda}\lambda^3}{3!} = \frac{e^{-\lambda}\lambda^4}{4!}$ Since $e^{-\lambda} > 0$ and $\lambda > 0$, we can divide both sides by $e^{-\lambda}\lambda^3$: $\frac{1}{3!} = \frac{\lambda}{4!}$ Multiplyi.

44 [Challenging] • Target: Graphing a parametric curve after elimination • Cross-Lesson: 4.01 Circles and Parabolas

The function f is defined by $f(x) = a \cdot b^x + c$, where a, b, and c are constants with $a > 0$, $b > 0$, and $beq1$. In the xy-plane, the horizontal asymptote of the graph of $y = f(x)$ is the line $y = -4$. If the graph passes through the points $(0, -1)$ and $(2, 8)$, what is the value of $f(3)$?

a. **20** ✓

b. 24

c. 32

d. 44

Step 1: Determine the vertical shift parameter c. For an exponential function in the form $f(x) = a \cdot b^x + c$, the term $a \cdot b^x$ approaches 0 as $x \to -\infty$ (for $b > 1$) or as $x \to \infty$ (for 00), we find $b = 2$. Step 4: Evaluate $f(3)$. The complete function is $f(x) = 3 \cdot 2^x - 4$. Evaluating at $x = 3$: $f(3) = 3 \cdot 2^3 - 4 = 3(8) - 4 = 24 - 4 = 20$ Misconception Traps: - Choice B (24) results from computing $a \cdot b^3 = 3 \cdot 8 = 24$ but forgetting to subt.

© ClipSAT — clipsat.org | All Rights Reserved

45 [Challenging] • Target: Restricting the parameter domain • Cross-Lesson: 4.02 Ellipses

A transformation is applied to a triangle ABC, mapping it to triangle A'B'C'. The transformation can be represented by a matrix multiplication of the form $M(x; y) = (3, 1; 1, k)(x; y)$, where M is the transformation matrix and k is an unknown parameter. If the original point A is $(2, 1)$ and its transformed image A' is $(7, 5)$, find the value of k.

a. 2

b. **3** ✓

c. 4

d. 5

The transformation matrix M is given by $M = (3, 1; 1, k)$. We are given that point A $(2, 1)$ is transformed to point A' $(7, 5)$. This can be expressed as a matrix equation: $M(2; 1) = (7; 5)$. Substituting the matrix M, we get: $(3, 1; 1, k)(2; 1) = (7; 5)$. Performing the matrix multiplication for the first row: $(3 \cdot 2) + (1 \cdot 1) = 6 + 1 = 7$. This matches the first component of A', confirming the top row of the matrix is corre.

© ClipSAT — clipsat.org | All Rights Reserved